

AI-IBED: AI-Supported Formative Assessment Rubrics

Teacher-ready assessment toolkit for AI_IBED

Purpose: Use observable student evidence to guide feedback during inquiry, engineering design and reflection. AI may suggest interpretations and questions; the teacher verifies evidence and makes all assessment decisions.

Performance scale

1 Beginning: substantial support needed • 2 Developing: partial demonstration • 3 Proficient: independent and accurate demonstration • 4 Advanced: integrated, critical and transferable demonstration. Use “Not assessed” when the task did not provide adequate evidence.

A. Scientific inquiry and conceptual understanding

Assessment evidence: Investigative questions, predictions, experimental plans, data sheets, models and explanations

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
Question formulation	Identifies a topic but cannot formulate an investigable question.	Develop a question with teacher support.	Formulates a clear, investigable scientific question.	Develops a focused question addressing relationships among multiple variables.
Scientific understanding	Shows major misconceptions or limited understanding.	Explain some relevant concepts but have gaps.	Accurately explains and applies relevant scientific concepts.	Integrates multiple concepts to explain and predict phenomena.
Investigation planning	Needs substantial support to identify variables and procedures.	Identifies some variables and a partially workable plan.	Designs a systematic investigation with appropriate variables and controls.	Justifies and refines the investigation considering reliability and alternatives.
Evidence interpretation	Describes observations without connecting them to concepts.	Identifies patterns but provides limited explanations.	Interprets evidence and connects findings to scientific concepts.	Critically evaluates evidence, uncertainty and alternative explanations.

AI-supported teacher prompt: Review this anonymized investigation plan. Identify evidence for each criterion, possible misconceptions and three probing questions. Do not assign a final grade.

B. Engineering design and iterative problem-solving

Assessment evidence: Design sketches, prototype plans, testing records, design revisions and engineering journals

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
Problem definition	Describes the problem without design requirements.	Identifies some criteria and constraints.	Defines the problem with measurable criteria and relevant constraints.	Analyzes competing criteria, stakeholder needs and constraints.
Solution development	Proposes a solution without a scientific rationale.	Proposes a feasible solution with limited justification.	Develops and justifies solutions using science and engineering.	Compare alternatives and justify decisions using evidence.
Prototype testing	Tests without systematically recording results.	conducts basic tests and records some results.	Uses systematic tests and interprets results against criteria.	Evaluates reliability, limitations and comparative tests.
Iterative improvement	Changes the design without explaining why.	Revises using limited testing evidence.	Uses testing evidence to justify improvements.	Conducts multiple evidence-based iterations and evaluates effects.

AI-supported teacher prompt: Compare anonymized initial and revised designs. Highlight changes supported by test evidence and ask questions about design trade-offs.

C. Systems thinking and sustainability reasoning

Assessment evidence: Concept maps, written justifications, systems diagrams and intervention proposals

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
System components	Identifies one or two isolated elements.	Identifies several relevant components.	Identifies components and explains system boundaries.	Justifies boundaries and considers overlooked components.
Relational reasoning	Describes components independently.	Identifies simple cause-and-effect relationships.	Explain multiple interconnected relationships.	Explain indirect relationships and cross-level interactions.
Feedback and temporal reasoning	Provides a linear explanation without considering time.	Recognizes a feedback relationship or change over time.	Explains feedback and system change over time.	Analyzes reinforcing/balancing feedback and long-term consequences.
Leverage points and trade-offs	Suggests an intervention without explaining effects.	Identifies an intervention and some consequences.	Justifies an intervention, considering benefits and trade-offs.	Compares interventions, unintended effects and stakeholder priorities.

AI-supported teacher prompt: Review anonymized successive concept maps and written justifications. Identify supported new relationships, feedback and temporal reasoning; flag unsupported interpretations.

D. Scientific argumentation and communication

Assessment evidence: Claim–evidence–reasoning responses, presentations, posters and peer discussions

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
Scientific claim	Makes an unclear or unsupported claim.	Makes a relevant claim needing clarification.	Presents a clear, scientifically appropriate claim.	Presents a precise, qualified claim accounting for complexity.
Evidence	Provides little or irrelevant evidence.	Provides some evidence without evaluating quality.	Selects appropriate evidence to support the claim.	Integrates multiple sources and evaluates limitations.
Reasoning	Does not connect evidence to the claim.	Partially connects evidence and claim.	Uses scientific principles to connect evidence and claim.	Addresses alternative explanations in a coherent argument.
Communication	Presents ideas that are difficult to follow.	Communicates main ideas with some gaps.	Communicates a logical, accessible scientific explanation.	Adapts to audience and responds using evidence.

AI-supported teacher prompt: Review the original CER response. Suggest clarity improvements and probing questions, while preserving the student’s own reasoning.

E. Collaboration, reflection and responsible AI use

Assessment evidence: Group observation, reflections, revision history and AI-use logs

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
Collaboration	Participates minimally or needs substantial support.	Contributes inconsistently.	Share responsibilities and responds constructively to peers.	Coordinates contributions and resolve disagreements using evidence.
Reflection	Describes activities without evaluating learning.	Identifies successes or difficulties with limited explanation.	Explains progress and identifies specific next steps.	Critically examines changing thinking and applies reflection.
AI literacy	Accepts AI information without checking it.	Recognizes possible errors but checks inconsistently.	Verifies AI output and explains its role in the task.	Critically evaluates limitations and documents responsible use.
Feedback use	Rarely responds to feedback.	Makes limited revisions following feedback.	Uses feedback to make purposeful improvements.	Evaluates multiple feedback sources and justifies revisions.

AI-supported teacher prompt: Review an anonymized reflection and AI-use log. Distinguish the student’s own reasoning from AI-generated suggestions; propose one actionable next step.

F. Real-world application and sustainable decision-making

Assessment evidence: Context analyses, solution proposals, life-cycle comparisons and action plans

Criterion	1 · Beginning	2 · Developing	3 · Proficient	4 · Advanced
Contextual understanding	Identifies a problem in general terms.	Describes the problem and some local effects.	Explains the problem with scientific and contextual evidence.	Integrates scientific, environmental, economic and social factors.
Solution feasibility	Proposes a solution without practical constraints.	Considers some implementation requirements.	Proposes a feasible solution and needed resources.	Evaluates feasibility across contexts and anticipates challenges.
Sustainability impacts	Identifies one environmental benefit.	Recognizes several benefits or drawbacks.	Evaluates impacts and relevant trade-offs.	Uses life-cycle reasoning to compare impacts and unintended effects.
Action and evaluation	Suggests an action without an implementation plan.	Proposes basic implementation steps.	Develops an actionable plan with measurable indicators.	Evaluates outcomes and proposes evidence-based improvements.

AI-supported teacher prompt: Review an anonymized sustainability proposal. Identify unsupported environmental claims, overlooked trade-offs and evidence needed to test feasibility.

Alignment with the AI-IBED instructional cycle

Engage and Ask — Diagnostic questions and prior knowledge | **Evidence:** Entry tickets, initial maps, problem statements.

Explore and Research — Inquiry planning and evidence collection | **Evidence:** Plans, research notes, experimental records.

Explain and Imagine — Conceptual explanation and alternative solutions | **Evidence:** Revised maps or solution sketches.

Elaborate, Plan and Create — Application and design justification | **Evidence:** Design proposals, prototypes, journals.

Test, Evaluate and Improve — Performance, trade-offs and iteration | **Evidence:** Test results, comparisons, revised prototypes.

Communicate and Reflect — Transfer, communication and reflection | **Evidence:** Presentations, posters, reflections, action plans.

Reusable AI feedback protocol

1. Collect anonymized student evidence.
2. Ask AI for criterion-specific, evidence-based provisional interpretations.
3. Verify against original work and classroom observations.
4. Give one strength, one improvement area and an achievable next step.
5. Reassess using new evidence.

Reusable prompt: *Analyze the anonymized student work using the relevant AI-IBED rubric. Cite observable evidence for each criterion, suggest a provisional performance level, distinguish missing evidence from misconceptions, generate three probing questions, and recommend one differentiated activity. Do not invent student responses or assign a final grade.*

Teacher assessment record (copy for each student)

Field	Teacher entry
Student identifier	
Lesson / project	
Assessment date	
Rubric dimension(s)	
Observed evidence	
Provisional level and rationale	
Feedback to student	
Next instructional step	
Reassessment evidence / date	

Illustrative application: Bioplastic design challenge

Student response: “Our starch-based plastic was flexible, but it tore when we added too much water. We think reducing the amount of water might make it stronger. Bioplastics are better for the environment because they break down.”

Possible formative interpretation:

1. Inquiry—Developing (observes a relationship but needs controlled testing)
2. Engineering design: Developing (proposes an untested revision)
3. Systems thinking—Beginning (environmental claim lacks consideration of disposal conditions)
4. Argumentation—Developing (observations are relevant, environmental claim lacks evidence).

Probing questions: How could you test whether water content changes strength? Under what conditions would this material break down? What other impacts should be considered before calling it more sustainable?

Implementation safeguards

These are proposed formative descriptors, not a validated measurement instrument. It only serves as a personal guide to making your lesson align to State’s standards. Follow district privacy policies; do not upload identifiable student work to unapproved AI services. Ensure that students can demonstrate understanding without AI assistance.